Mechanical Design Report

Final Project 2026 — Delivery Robot

Alexandria University Robotics (AUR) | Mechanical Team

1 Introduction

The task of this year's final project is to build a small robot that picks up cardboard boxes from behind a 3 cm wall and delivers each one to its destination on a 3×3 m playground. Each box is $6 \times 6 \times 6$ cm. The robot must also stay inside the size limit of $30 \times 30 \times 40$ cm.

This report explains the mechanical side of our robot: the body, the wheels, the gripper, and the space we kept for the other teams' parts. It also explains why we picked each design and what the other options were.

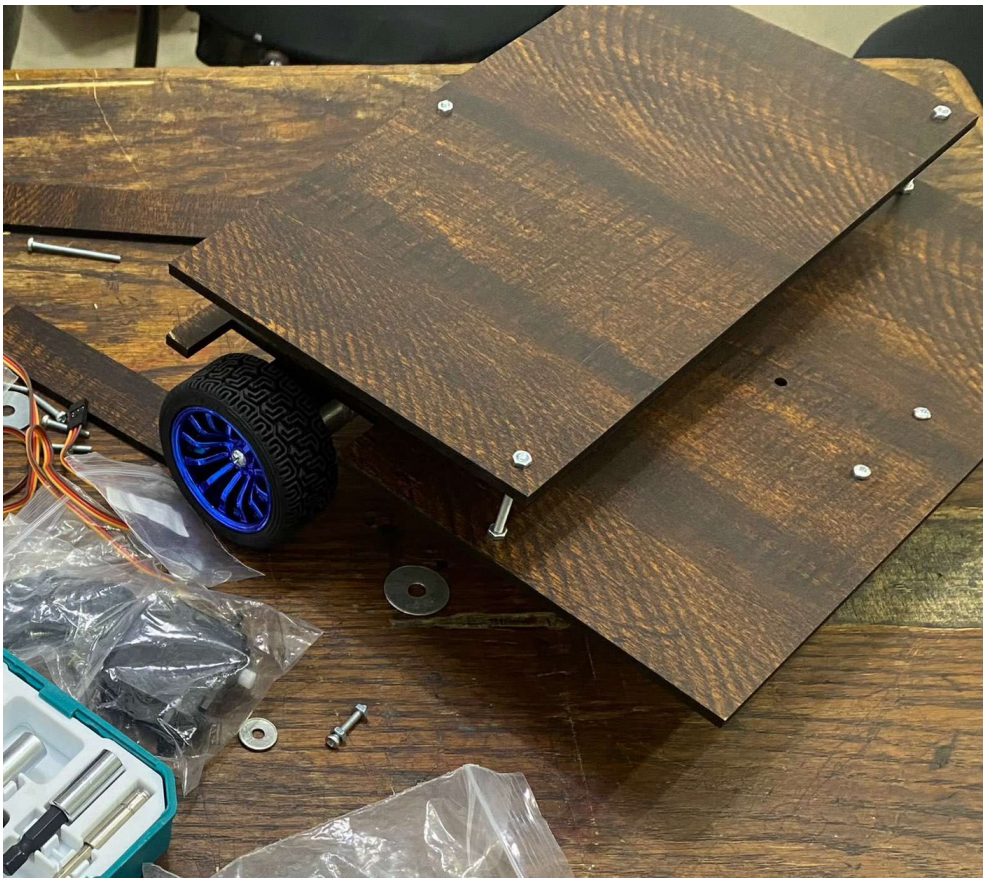


Figure 1: The robot body during assembly. The two levels are laser-cut wood, joined with bolts, nuts and spacers.

2 Design Goals

We started from the rulebook and from what the other subteams told us they need. Our goals were:

- Stay inside the size limit of 30 cm long, 30 cm wide and 40 cm high.
- Hold the boxes firmly while the robot drives, so they do not fall.
- Keep the robot level and stable, so the sensors and camera read correctly.
- Leave enough room for the boards, battery, wiring and sensors.
- Make the parts quickly and cheaply, because we have only a short time before competition day and a limited budget.
- Make the robot easy to open and fix.

3 Design Process

We worked in four steps.

1. **Sketching and meeting the other teams.** We talked with the hardware team about the size of their boards and the places where the sensors must sit.
2. **CAD design in SolidWorks.** We modelled both levels, the wheel mounts, the brackets and the gripper. This let us check that every part fits before cutting anything.
3. **Manufacturing.** We cut the wooden levels with a laser cutter and 3D printed the gripper.
4. **Assembly and checking.** We assembled the robot, levelled it and tested that everything is firm.

4 Chassis

4.1 Two-level body

The body has two levels made of laser-cut wood. The **lower level (base)** is much larger than the upper level. We did this on purpose:

- The base carries the wheels, the motors and their brackets, so it needs the extra room around its edges.
- The smaller upper level sits inside the base area. This leaves free space on the base for the gripper and for the box to be placed and carried.
- Splitting the parts over two levels uses the height (up to 40 cm) instead of making the robot longer or wider.

The two levels are held apart by bolts, nuts and spacers. The upper level carries the electronics and the base carries the drive system. Putting the heavier motors low also keeps the centre of gravity low, which helps the robot stay stable when it stops and turns.

4.2 Why wood

We chose laser-cut wood because it is less costly than the other options and it gives accurate flat plates in a short time. Wood sheets are cheap, and the laser cut wastes very little material, which helps us stay under the 3500 EGP budget cap. The laser cut also gives exact holes and slots straight from the SolidWorks drawing, so the parts fit on the first try. We kept 3D printing for the gripper, which has a complex shape.

Option	Good	Not so good
Laser-cut wood	Lowest cost, fast, flat and accurate, easy to drill and bolt	Heavier than plastic, can chip if over-tightened
3D printing	Any shape is possible	Slow for large flat plates, uses a lot of material
Acrylic	Clean look, accurate	Can crack around bolt holes, costs more
Aluminium sheet	Strong and light	Needs bending and drilling tools, costs more

Table 1: Material options for the body.

5 Drive System

5.1 Wheel layout

The robot uses **two drive wheels at the back** and **one caster wheel at the front**. Each drive wheel has its own DC motor, so the robot turns by running the two motors at different speeds.

- **Turns in a small space:** the robot can rotate almost on the spot, which is useful in a small playground and near the boxes.
- **Stable:** three contact points always touch the ground, so no suspension is needed.

The other option was four wheels. It gives more grip but it needs more motors and drivers, costs more, and makes turning harder because the wheels drag sideways. It was not worth it for this task.

5.2 Motor mounting

The DC motors are bolted straight to the body through brackets. The bolts we had were too small for the bracket holes, so we added **washers** to spread the load and hold the brackets firmly. Space for the wheels and the brackets was planned in the CAD, so the wheels do not touch the body.

5.3 Motor specifications

Both drive wheels use the **GA25-370 DC gear motor with encoder**. It comes with a metal mounting bracket, which we bolt to the body. The encoder gives feedback on wheel speed and distance for the robot's motion control.

Parameter	Value
Product type	DC geared motor with encoder
Motor model	GA25-370
Motor type	Brushed DC motor
Rated voltage	12 V DC
Rated speed	130 RPM
Rated torque	4.4 kg.cm
Gearbox	Metal gearbox
Encoder	Incremental quadrature, channels A and B
Rotation direction	CW / CCW
Output shaft	D-type
Mounting	Metal bracket included
Housing material	Metal

Table 2: Specifications of the drive motor.

5.4 Levelling

For the robot to drive straight, the body must be level. We used spacers between the parts to adjust the height of each side, and we checked the result with a water level. We repeated this until the bubble stayed in the middle. A level body also keeps the camera pointing in the right direction and the sensors reading correctly.

6 Gripper

6.1 Design

The gripper must hold a $6 \times 6 \times 6$ cm cardboard box without crushing it. We designed it in SolidWorks and **3D printed** it. It is driven by **two servo motors** (MG996R):

- **Servo 1** opens and closes the two jaws through a linkage. Both jaws move together, so the box ends up in the centre of the gripper.
- **Servo 2** moves the gripper, so it can lift and lower the box.

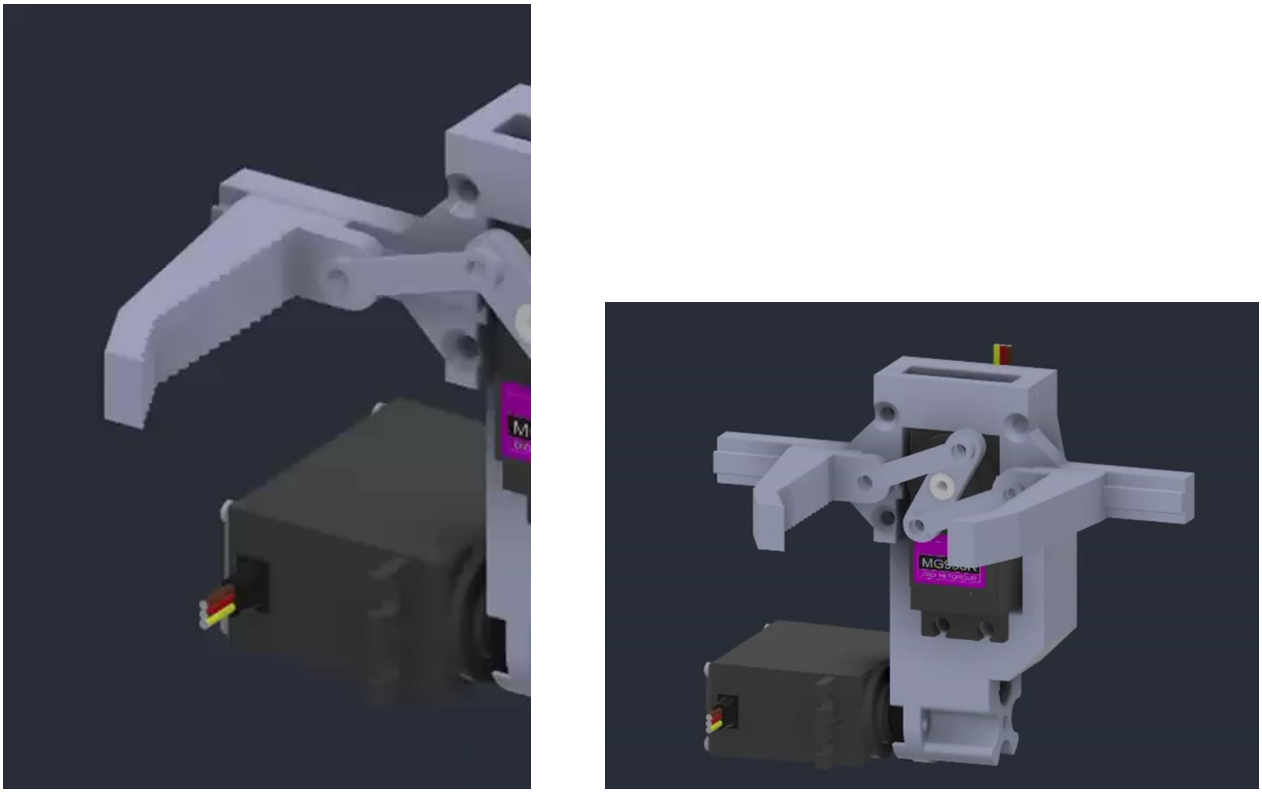


Figure 2: SolidWorks model of the gripper: the jaws, the linkage and the two servos.

6.2 Why this design

- **Centring:** two jaws moving together push the box to the middle, so small errors in the robot's position do not matter much.
- **Strong enough:** the MG996R is a high-torque servo, which is more than enough for a light cardboard box.
- **Easy to print:** the parts are simple, so they print without support problems, and we can reprint any broken part quickly.
- **Low cost:** servos are cheap and the plastic used is little.

The gripper is mounted at the front of the robot, so it can reach the box from behind the 3 cm wall.

7 Space for the Other Subteams

We talked with the hardware team before cutting the wood. We took the size of their boards, the battery and the wiring, and we left room for them on the upper level. We also left space for the sensors in the places they asked for. Because this was done in the CAD stage, we did not need to change the body afterwards.

8 Dimensions

Dimension	Rulebook limit	Our robot
Length	30 cm	27 cm
Width	30 cm	23 cm
Height	40 cm	_____ cm

Table 3: Robot size compared to the rulebook limit.

9 Problems and Fixes

- **Motor bolts too small for the bracket holes.** We added washers.
- **Robot not level.** We added spacers and checked with a water level.
- **Not enough room for the electronics.** We planned this in CAD together with the hardware team, so the body was made right the first time.

10 Conclusion

We built a two-level wooden robot with two drive wheels, a front caster, and a servo-driven 3D printed gripper. The design is simple, cheap and quick to make, it fits the size limit, and it leaves space for the hardware and software teams. If we had more time, we would test lighter wood or a lighter material for the upper level, and add stronger grip pads to the gripper jaws.